



COMPETENCY STANDARD

FOR

HEAVY EQUIPMENT OPERATION
(ROAD ROLLER & EXCAVATOR)

(CONSTRUCTION SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Heavy Equipment Operation (Road Roller & Excavator) serves as a foundational document for creating curricula, teaching and learning materials, and assessment tools. It also facilitates training aligned with industry requirements, ensuring that individuals who meet the set standards through assessment are qualified and ready for relevant job roles.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been reviewed and updated by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:**Generic Competencies (30 Hrs.)**

SL	Code	Unit of competency	Elements of competency	Duration (hours)
1.	SICIP-CON-RRE-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
2.	SICIP-CON-RRE-02-G	Work in a team environment	1. Identify team goals and processes 2. Communicate and cooperate with team members 3. Work as a team member 4. Solve problems as a team member	20
	Total hours			30 Hrs.

Occupation Specific Competencies (330 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-CON-RRE-01-O	Identify Basic Concepts of Heavy Equipment Operation (Road Roller & Excavator)	1. Identify machine purpose and functions. 2. Identify safety rules of Heavy Equipment Operation (Road Roller & Excavator). 3. Describe key parts and components. 4. Explain basic operating principles of roller. 5. Explain basic operating principles of excavator. 6. Identify hand tools and power tools.	30
SICIP-CON-RRE-02-O	Inspect and prepare road roller & excavator	1. Check machine body and safety guards. 2. Check oil, coolant, fuel, greasing and hydraulic levels. 3. Check brakes, lights, horn, engine smoke and alarms. 4. Check tires, drums and undercarriage. 5. Check controls, gauges and systems.	20
SICIP-CON-RRE-03-O	Operate Road Roller	1. Start and control the roller. 2. Perform the roller on various site conditions. 3. Operate roller safely on slopes. 4. Load and unload road roller (Trailer) safely. 5. Park and shut down safely.	130
SICIP-CON-RRE-04-O	Operate Excavator	1. Start and position the excavator for tasks. 2. Control boom, arm, bucket, blade and swing smoothly. 3. Perform digging, trenching, lifting, and loading. 4. Load and unload excavator (Trailer) safely. 5. Follow safe travel, turning, and parking steps.	130

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-CON-RRE-05-O	Maintain and Care of Road Roller and Excavator	1. Clean machinery after operation. 2. Lubricate moving parts routinely. 3. Inspect components for wear or damage. 4. Perform routine maintenance. 5. Store machinery and tools safely.	20
Total Hours			330 Hrs.

COMPETENCY STANDARD: HEAVY EQUIPMENT OPERATION (ROAD ROLLER & EXCAVATOR)

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-RRE-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OH&S) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International OHS requirements 1.2 Bangladesh standards for OHS

	1.3 Building Code 1.4 Fire Safety Rules and Regulations 1.5 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gas Mask 2.3 Gloves 2.4 Safety shoes 2.5 Helmet 2.6 Face mask 2.7 Overalls 2.8 Goggles and safety glasses 2.9 Ear plugs 2.10 Sun block 2.11 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards. 3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Tourniquet 5.5 Applying CPR (Cardiopulmonary Resuscitation) 5.6 Taking appropriate medicine

Curricular Evidence Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices

	2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Performing first aid. 2.7 Performing basic firefighting accessories using fire extinguishers 2.8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace. 3.3 Promptness in carrying out activities 3.4 Tidiness and timeliness 3.5 Respect of peers, sub-ordinates and seniors in workplace 3.6 Environmental concern 3.7 Sincere and honest to duties
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Reported incidents arising from hazards and risks to authority. 1.4 Emergency response plans and procedures are implemented. 1.5 Applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK IN A TEAM ENVIRONMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-RRE-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal <u>forms of communication</u> are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood and valued. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications and workplace procedures. 3.3 Team member's support with other members are made to ensure team achieves goals, awareness and requirements. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of variables:

Variables	Range (may include but not limited to)
1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning. 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Support mechanism to other members of team to ensure achievements of goals. 1.10 Methods of identifying current and potential problems faced by a team 1.11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures. 2.8 Supporting other members of the team to ensure team achieves goals, awareness and requirements. 2.9 Identifying current and potential problems faced by the team 2.10 Identifying solutions to the problem 2.11 Solving problems effectively and evaluating the outcome of the implemented solution

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Work books

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 communicated and cooperated with team members. 1.3 worked as a team member 1.4 solved problems as a team member
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: IDENTIFY BASIC CONCEPTS OF HEAVY EQUIPMENT OPERATION (ROAD ROLLER & EXCAVATOR)	Nominal Duration: 30 Hrs.	Unit Code: SICIP-CON-RRE-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to identify basic concepts of Heavy Equipment Operation (Road Roller & Excavator). It specifically includes the tasks of identifying machine purpose and functions, identifying safety rules of heavy equipment & excavator, describing key parts and components, explaining basic operating principles of roller, explaining basic operating principles of excavators and identifying hand tools and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify machine purpose and functions.	1.1 <u>Types of road rollers</u> are described based on design and application. 1.2 <u>Types of excavators</u> are described according to structure and usage. 1.3 Purpose and basic functions of road rollers and excavators are identified and explained. 1.4 Key features and operational capabilities of each machine type are recognized. 1.5 Differences between various models and configurations are identified.
2. Identify safety rules of Heavy Equipment Operation (Road Roller & Excavator).	2.1 <u>Personal protective equipment (PPE)</u> requirements are recognized. 2.2 <u>Safety rules for road rollers</u> are identified and understood. 2.3 <u>Safety rules for excavators</u> are identified and understood. 2.4 <u>Potential hazards</u> related to machine operation are identified through risk assessment. 2.5 Emergency procedures and safe shutdown steps are explained and followed.
3. Describe key parts and components.	3.3 <u>Key parts and components of road rollers</u> are identified. 3.4 <u>Key parts and components of excavators</u> are identified. 3.5 Functions of major components of road roller and excavator are described. 3.6 Hydraulic, mechanical, and electrical systems are

	explained. 3.7 Operator controls and dashboard indicators are recognized and interpreted.
4. Explain basic operating principles of roller.	4.1 Role of vibration, static weight, and compaction force is described. 4.2 Forward and reverse movement controls are interpreted and understood. 4.3 Steering, braking, and speed control mechanisms are explained. 4.4 Compaction patterns and passes are described for effective operation. 4.5 Soil type and moisture are recognized.
5. Explain basic operating principles of excavator.	5.1 Boom, arm, cylinder and bucket movement functions are described. 5.2 Hydraulic system operations that power machine movements are interpreted. 5.3 Swing, travel, and lifting mechanisms are explained. 5.4 Controls for digging, loading, and dumping are understood. 5.5 Ground stability and load limits are recognized.
6. Identify hand tools and power tools.	6.1 <u>Hand tools used for road roller and excavator</u> are identified. 6.2 <u>Power tools</u> required for servicing and adjustments are recognized. 6.3 Functions and uses of each tool are described.

Range of Variables

Variable	Range (Includes but not limited to):
1. Types of road rollers	1.1 Soil compactor 1.2 Tandem roller 1.3 Pneumatic Tire Roller 1.4 Static Roller
2. Types of excavators	2.1 Crawler Excavator 2.2 Wheeled Excavator 2.3 Long-Reach Excavator 2.4 Amphibious Excavator
3. Personal protective equipment (PPE)	3.1 Helmet 3.2 Apron 3.3 Goggles 3.4 Face mask 3.5 Ear plug 3.6 Hand gloves 3.7 Safety shoes
4. Safety rules for road rollers	4.1 Wear PPE 4.2 Inspect the machine 4.3 Check brakes, lights, horn

	4.4 Keep area clear 4.5 Use seatbelt 4.6 Control travel speed and slopes 4.7 Avoid sharp turns on slopes 4.8 Stay away from edges 4.9 Stop in poor visibility 4.10 Park on level ground
5. Safety rules for excavators	5.1 Wear PPE 5.2 Inspect the machine 5.3 Check controls and hydraulics 5.4 Keep bystanders away 5.5 Use seatbelt 5.6 Operate at safe speed 5.7 Avoid swinging over people or vehicles 5.8 Stay clear of edges and trenches 5.9 Use proper signals with spotters 5.10 Stop work in poor visibility 5.11 Park on level ground
6. Potential hazards	6.1 Fire hazards 6.2 Electrical hazards 6.3 Mechanical hazards 6.4 Chemical hazards 6.5 Environmental hazards 6.6 Falling object hazards 6.7 Ground collapse hazards 6.8 Visibility hazards 6.9 Noise and vibration hazards
7. Key parts and components of road rollers	7.1 Drum (Front/Rear) 7.2 Engine 7.3 Vibratory system 7.4 Steering and control panel 7.5 Hydraulic system 7.6 Operator's cabin / seat system
8. Key parts and components of excavators	8.1 Boom 8.2 Arm (Stick) 8.3 Cylinder 8.4 Swing system 8.5 Tracks or wheels 8.6 Travel motor 8.7 Lever 8.8 Hydraulic system 8.9 Engine 8.10 Operator cabin
9. Hand tools used for road roller and excavator	9.1 Spanner set 9.2 Socket or ratchet set 9.3 Screwdrivers (Flat and Star) 9.4 Pliers

	9.5 Allen key set 9.6 Hammer 9.7 Wrenches 9.8 Filter opener 9.9 Grease gun 9.10 Measuring tape 9.11 Cleaning brush
10. Power tools	10.1 Impact wrench 10.2 Electric drill 10.3 Hand grinder 10.4 Blower 10.5 Air compressor 10.6 Pressure washer 10.7 Portable lighting set 10.8 Battery-operated grease gun

Curricular Content Guide

1. Underpinning Knowledge	1.1. Concept of different types of road rollers and excavators. 1.2. Purpose and fundamental functions of road rollers and excavators in construction and earth-moving work. 1.3. Key features, capabilities, and differences among various machine models and configurations. 1.4. Major parts and components of road rollers and excavators. 1.5. Basic operating principles of road rollers. 1.6. Basic operating principles of excavators. 1.7. Common hand tools and power tools used for servicing heavy equipment and the basic purpose of each tool.
2. Underpinning Skills	2.1. Identifying machine types, features, and functions. 2.2. Interpreting safety rules, PPE use, and hazards. 2.3. Recognizing parts, components, and system functions. 2.4. Interpreting controls, indicators, and operating mechanisms. 2.5. Explaining basic operating principles of roller and excavator. 2.6. Identifying and selecting hand tools and power tools. 2.7. Following safe operating and emergency procedures.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace

4 Resource Implications	The following resources must be provided: 4.1 Workplace (actual or simulated) 4.2 Tools, equipment and facilities appropriate to processes or activity 4.3 Materials relevant to the proposed activity
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified machine purpose and functions. 1.2 identified safety rules of Heavy Equipment Operation (Road Roller & Excavator). 1.3 described key parts and components. 1.4 explained basic operating principles of roller. 1.5 explained basic operating principles of excavator. 1.6 identified hand tools and power tools.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INSPECT AND PREPARE ROAD ROLLER & EXCAVATOR	Nominal Duration: 20 Hrs.	Unit Code: SICIP-CON-RRE-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to inspect and prepare road roller & excavator. It specifically includes the tasks of checking machine body and safety guards, checking oil, coolant, fuel, greasing and hydraulic levels, checking brakes, lights, horn, engine smoke and alarms, checking tires, drums and undercarriage and checking controls, gauges and systems.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check machine body and safety guards.	1.1 <u>Machine body for road roller</u> is inspected for cracks, dents, corrosion, or structural damage. 1.2 <u>Machine body for excavator</u> is inspected for cracks, dents, corrosion, or structural damage. 1.3 Exterior panels and covers are checked for proper fitting and condition. 1.4 Safety guards and protective shields are checked to ensure they are secure and intact. 1.5 Missing, loose, or damaged guards are identified. 1.6 Warning labels and safety decals are checked for visibility and readability.
2. Check oil, coolant, fuel, greasing and hydraulic levels.	2.1 Engine oil level is checked and compared with recommended limits. 2.2 Coolant level is inspected to ensure proper temperature control. 2.3 Fuel level is verified for sufficient operation time as per manufacturing requirements. 2.4 Greasing points are checked to confirm lubrication is adequate. 2.5 Hydraulic oil level is inspected for safe and smooth machine movement. 2.6 Leaks or abnormal fluid conditions are identified and reported.
3. Check brakes, lights, horn, engine smoke and alarms.	3.1 Braking system is inspected to ensure proper functionality. 3.2 Foot brakes and parking brakes are tested for responsiveness. 3.3 All working lights, including headlights and indicators, are checked for operation. 3.4 Horn is tested to confirm it produces a clear and audible sound. 3.5 Safety alarms and backup warning signals are tested for

	proper activation. 3.6 Engine smoke is observed and assessed for abnormal color such as white or black.
4. Check tires, drums and undercarriage.	4.1 Tires are inspected for cuts, wear, pressure, and proper inflation. 4.2 Roller drums are checked for cracks, dents, and surface damage. 4.3 Drum vibration components are reviewed for proper condition. 4.4 <u>Undercarriage parts</u> are checked for wear. 4.5 Loose, damaged, or misaligned components are identified and reported.
5. Check controls, gauges and systems.	5.1 <u>Operator controls</u> are tested to ensure smooth and correct response. 5.2 <u>Gauges and hydraulic indicators</u> are checked for accuracy. 5.3 Warning lights and display messages are observed for proper function. 5.4 Control levers, pedals, and switches are inspected for wear or malfunction. 5.5 <u>Hydraulic and electrical systems</u> are tested for proper operation. 5.6 Irregular readings or system faults are identified and reported.

Range of Variables

Variable	Range (Includes but not limited to):
1. Machine body for road roller	1.1 Operator cabin / seat area 1.2 Control panel 1.3 Engine compartment / engine cover 1.4 Main frame / chassis 1.5 Front drum (roller drum) 1.6 Rear drum or wheels (depending on type) 1.7 Water tank and sprinkler system (if fitted) 1.8 ROPS / canopy and safety guards
2. Machine body for excavator	2.1 Cab / Operator Cabin 2.2 Engine Compartment 2.3 Upper Frame / Upper Structure 2.4 Counterweight 2.5 Boom 2.6 Arm / Dipper 2.7 Bucket 2.8 Cylinder 2.9 Undercarriage
3. Undercarriage parts	3.1 Swing system 3.2 Track shoe 3.3 Travel motor

	3.4 Sprocket 3.5 Carrier roller and idler roller
4. Operator controls	Excavator 4.1 Joystick controls 4.1.1 Travel liver 4.1.2 Swing liver 4.1.3 Arm liver 4.1.4 Boom liver 4.2 Throttle control 4.3 Blade control 4.4 Monitor panel 4.5 Emergency stop Road Roller 4.6 Steering control 4.7 Drive control 4.8 Brake control 4.9 Vibration control 4.10 Water spray control
5. Gauges and hydraulic indicators	5.1 Engine oil gauges/indicator 5.2 Fuel gauge/indicator 5.3 Battery gauge/indicator 5.4 Pressure gauge 5.5 Temperature gauge 5.6 Hydraulic level indicator 5.7 RPM 5.8 Hour meter
6. Hydraulic and electrical systems	Hydraulic System 6.1 Hydraulic pump 6.2 Control valve 6.3 Hydraulic cylinders 6.4 Hydraulic hoses 6.5 Hydraulic tank Electrical System 6.6 Battery 6.7 Starter motor 6.8 Alternator 6.9 Fuses and relays 6.10 Wiring harness

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic concept of machine body inspection for detecting cracks, dents, corrosion, and structural damage in road rollers and excavators. 1.2 Purpose of safety guards, shields, and warning labels, and the need. 1.3 Key ideas behind monitoring essential fluid levels.
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	1.4 Understanding common signs of leaks or abnormal fluid conditions. 1.5 Basic operating concept of braking systems, working lights, horns, alarms, and smoke indicators. 1.6 Knowledge of tire condition, drum surface condition, and undercarriage components related to wear and alignment. 1.7 Concept of vibration components in road rollers and why their condition affects compaction. 1.8 Basic idea of hydraulic and electrical system functions.
2. Underpinning Skills	2.1 Inspecting machine body, panels, and safety guards. 2.2 Checking engine oil, coolant, fuel, greasing, and hydraulic levels. 2.3 Testing brakes, lights, horn, alarms, and monitoring engine smoke. 2.4 Inspecting tires, drums, vibration parts, and undercarriage components. 2.5 Testing controls, gauges, indicators, and system functions. 2.6 Identifying faults, irregularities, leaks, or damaged parts. 2.7 Reporting defects and unsafe conditions accurately.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	The following resources must be provided: 4.1 Workplace (actual or simulated) 4.2 Tools, equipment and facilities appropriate to processes or activity 4.3 Safety guideline

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 checked machine body and safety guards. 1.2 checked oil, coolant, fuel, greasing and hydraulic levels. 1.3 checked brakes, lights, horn, engine smoke and alarms. 1.4 checked tires, drums and undercarriage. 1.5 checked controls, gauges and systems.
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2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE ROAD ROLLER	Nominal Duration: 130 Hrs.	Unit Code: SICIP-CON-RRE-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate road roller. It specifically includes the tasks of starting and controlling the roller, performing the roller on various site conditions, operating roller safely on slopes, loading and unloading road roller (trailer) safely and parking and shut down safely.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Start and control the roller.	1.1 <u>Pre-start checks</u> are reviewed before starting the roller. 1.2 Engine is started following the manufacturer's recommended procedure. 1.3 Controls are operated smoothly to move the roller forward and backward. 1.4 Steering is controlled to maintain a safe and steady path. 1.5 Speed is adjusted according to site conditions and compaction requirements. 1.6 Brake and stopping functions are applied safely when needed.
2. Perform the roller on various site conditions.	2.1 <u>Site conditions</u> are assessed before operation. 2.2 Roller movement is adjusted to match ground conditions and compaction needs. 2.3 Appropriate compaction patterns and passes are applied on different surfaces. 2.4 Speed and vibration settings are controlled to avoid over-compaction or machine damage. 2.5 Uneven, soft, or unstable areas are navigated safely.
3. Operate roller safely on slopes.	3.1 Slope angle and ground stability are assessed before operation. 3.2 Roller is positioned correctly at the top or bottom of the slope as required. 3.3 Speed is reduced to maintain control on inclines and declines. 3.4 Steering is controlled smoothly to prevent sudden movement or slipping. 3.5 Vibration settings are adjusted or turned off when unsafe on steep slopes. 3.6 Side-to-side rolling across steep slopes is avoided to reduce rollover risk.
4. Load and unload road roller (Trailer) safely.	4.1 Trailer condition, ramps, and securing points are inspected before loading or unloading. 4.2 Road roller is positioned and aligned correctly with the

	trailer ramps. 4.3 Speed is controlled slowly while driving the roller onto or off the trailer. 4.4 Spotters or hand signals are followed to ensure safe guidance during loading and unloading. 4.5 Roller is parked in the correct position on the trailer platform. 4.6 Wheel chocks, chains, and tie-downs are applied to secure the roller safely during transport. 4.7 All steps of unloading are performed carefully to prevent slips, tipping, or equipment damage.
5. Park and shut down safely.	5.1 A safe and level parking area is identified before shutting down the roller. 5.2 Roller is brought to a complete stop using proper braking procedures. 5.3 Parking brake is applied to secure the machine. 5.4 Engine is idled briefly to stabilize temperature before shutdown. 5.5 Engine is switched off following manufacturer guidelines. 5.6 Controls are reset to neutral and all switches are turned off. 5.7 Post-operation checks are performed and log book is maintained.

Range of Variables

Variable	Range (Includes but not limited to):
1. Pre-start checks	1.1 Master key on 1.2 Seat belt 1.3 Ignition switch
2. Site conditions	2.1 Soil type 2.2 Moisture 2.3 Slope
3. Post-operation checks	3.1 Clean the machine 3.2 Inspect for leaks 3.3 Check drum condition 3.4 Check fluid levels 3.5 Switch off controls 3.6 Secure parking brake 3.7 Shut down engine

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of pre-start inspection requirements and the purpose of checking a roller before operation. 1.2 Basic understanding of roller engine start-up principles and manufacturer-recommended starting procedures. 1.3 Key ideas behind forward/reverse movement, steering
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	control, speed adjustment, and braking functions of a road roller. 1.4 Understanding how soil type, moisture, stability, and surface condition influence compaction quality. 1.5 Basic principles of operating a roller on slopes. 1.6 Concept of safe loading and unloading practices. 1.7 Basic understanding of safe parking and shutdown requirements.
2. Underpinning Skills	2.1 Starting and controlling the roller smoothly and safely. 2.2 Adjusting roller movement based on different site conditions. 2.3 Operating the roller safely on slopes and uneven ground. 2.4 Applying correct compaction patterns, passes, and vibration settings. 2.5 Loading and unloading the roller on a trailer using safe procedures. 2.6 Parking, securing, and shutting down the roller properly. 2.7 Following spotter signals and maintaining safety throughout all operations.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (actual or simulated) 4.2 Tools, equipment and facilities appropriate to processes or activity. 4.3 Relevant manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 started and controlled the roller. 1.2 performed the roller on various site conditions. 1.3 operated roller safely on slopes. 1.4 loaded and unloaded road roller (Trailer) safely. 1.5 parked and shut down safely.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test

	2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE EXCAVATOR	Nominal Duration: 130 Hrs.	Unit Code: SICIP-CON-RRE-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate excavator. It specifically includes the tasks of starting and positioning the excavator for tasks, controlling boom, arm, bucket, blade and swing smoothly, performing digging, trenching, lifting and loading, loading and unloading excavator (trailer) safely and following safe travel, turning and parking steps.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Start and position the excavator for tasks.	1.1 <u>Pre-start checks</u> are reviewed before starting the excavator. 1.2 Engine is started following manufacturer-recommended procedures. 1.3 Excavator functions are tested to ensure proper operation. 1.4 Machine is positioned correctly according to the task requirements. 1.5 Ground stability and surrounding hazards are assessed before positioning.
2. Control boom, arm, bucket, blade and swing smoothly.	2.1 Boom, arm, bucket, blade and swing controls are operated smoothly and steadily. 2.2 Movements are adjusted according to task requirements and site conditions. 2.3 Jerky or abrupt motions are avoided to ensure safe operation. 2.4 Load handling is performed within the machine's capacity limits. 2.5 Control responses are monitored for accuracy and stability. 2.6 Surrounding obstacles and personnel are avoided during movement.
3. Perform digging, trenching, lifting, and loading.	3.1 Excavator movements are controlled to carry out digging, trenching, lifting, and loading tasks safely. 3.2 Bucket angle is adjusted to maintain efficient penetration and material retention. 3.3 Excavation force is managed to avoid machine strain, overloading, or ground collapse. 3.4 Depth, reach, and loading height are controlled according to task requirements. 3.5 Materials are placed, lifted, or loaded into designated areas without spillage.

	3.6 Safe distance from people, equipment, and utilities is maintained throughout the operation.
4. Load and unload excavator (Trailer) safely.	4.1 Trailer, ramps, and securing points are inspected before beginning loading or unloading operations. 4.2 Excavator is aligned accurately with the trailer ramps for safe entry or exit. 4.3 Speed is kept slow and controlled while driving the excavator onto or off the trailer. 4.4 Spotters or signaling personnel are followed to maintain clear and safe guidance. 4.5 Excavator is positioned correctly and evenly on the trailer platform. 4.6 Tie-downs, chains, and locking devices are applied securely to hold the machine in place. 4.7 Unloading steps are carried out carefully to prevent tipping, sliding, or equipment damage.
5. Follow safe travel, turning, and parking steps.	5.1 Travel path is inspected and cleared before moving the excavator. 5.2 Machine is driven at a controlled speed suitable for site conditions. 5.3 Boom and bucket are kept low and stable during travel for safety. 5.4 Turns are made slowly while maintaining balance and ground stability. 5.5 Obstacles, slopes, and uneven surfaces are avoided during movement. 5.6 Parking brake is applied, controls are set to neutral, and the machine is shut down safely. 5.7 <u>Post-operation checks</u> are performed and log book is maintained.

Range of Variables

Variable	Range (Includes but not limited to):
1. Pre-start checks	1.1 Walk-around machine check 1.2 Check engine oil and coolant 1.3 Check hydraulic oil and fuel level 1.4 Inspect tracks and undercarriage 1.5 Check boom, arm, and bucket 1.6 Check brake, lights, horn, and alarms 1.7 Check seat belt and controls
2. Post-operation checks	2.1 Park and lower attachments 2.2 Shut down engine 2.3 Check for leaks and damage 2.4 Clean machine 2.5 Inspect hydraulics 2.6 Check fluids 2.7 Report faults

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concept of pre-start inspection requirements for excavators. 1.2 Basic understanding of engine start-up principles and manufacturer. 1.3 Key operational functions of an excavator. 1.4 Basic concepts behind smooth operation of boom, arm, bucket, blade, and swing controls. 1.5 Understanding load limits, safe movement adjustment. 1.6 Concept of safe loading and unloading on trailers. 1.7 Basic understanding of safe travel practices-path inspection, controlled speed, low boom position. 1.8 Key ideas related to safe parking, shutdown requirements, and post-operation checks.
2. Underpinning Skills	2.1 Starting, positioning, and preparing the excavator for tasks. 2.2 Controlling boom, arm, bucket, blade, and swing smoothly. 2.3 Performing digging, trenching, lifting, and loading safely. 2.4 Adjusting movements based on task needs and site conditions. 2.5 Loading and unloading the excavator on a trailer safely. 2.6 Following safe travel, turning, parking, and shutdown procedures. 2.7 Maintaining awareness of surroundings, obstacles, and personnel at all times.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	The following resources must be provided: 4.1 Workplace (actual or simulated) 4.2 Tools, equipment and facilities appropriate to processes or activity 4.3 Materials relevant to the proposed activity 4.4 Relevant manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 started and position the excavator for tasks. 1.2 controlled boom, arm, bucket, blade and swing smoothly. 1.3 performed digging, trenching, lifting, and loading. 1.4 loaded and unloaded excavator (Trailer) safely. 1.5 followed safe travel, turning, and parking steps.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MAINTAIN AND CARE OF ROAD ROLLER AND EXCAVATOR	Nominal Duration: 20 Hrs.	Unit Code: SICIP-CON-RRE-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to maintain and care of road roller and excavator. It specifically includes the tasks of cleaning machinery after operation, lubricating moving parts routinely, inspecting components for wear or damage, performing routing maintenance and storing machinery and tools safely.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Clean machinery after operation.	1.1 <u>Cleaning tools and chemicals</u> are selected and used according to guidelines. 1.2 Machine surfaces are cleaned to remove dirt, mud, and debris. 1.3 Cab, controls, and windows are wiped and cleared for visibility. 1.4 Drums, buckets, and undercarriage areas are cleaned properly. 1.5 Cleaning activities are documented for maintenance records.
2. Lubricate moving parts routinely.	2.1 Lubrication points are identified according to the maintenance manual. 2.2 Grease and lubricants are applied to required joints and moving parts. 2.3 Excess grease or oil is removed for safety and cleanliness. 2.4 Lubrication intervals are followed as per manufacturer recommendations. 2.5 Lubrication activities are recorded for tracking.
3. Inspect components for wear or damage.	3.1 Critical components are inspected. 3.2 Signs of cracks, leaks, wear, or deformation are identified. 3.3 Hydraulic and electrical systems are checked for abnormalities. 3.4 Bolts, fittings, and fasteners are inspected for tightness. 3.5 Damaged or worn parts are reported immediately. 3.6 Unsafe components are tagged and isolated if necessary.
4. Perform routine maintenance.	4.1 Maintenance tools and equipment are used safely. 4.2 <u>Maintenance tasks</u> are carried out according to the schedule and manual.

	4.3 Calibration and adjustments are made to ensure proper function. 4.4 Minor repairs are performed within operator limits. 4.5 Major repair needs are reported to supervisors or technicians. 4.6 Maintenance logs are updated after each task.
5. Store machinery and tools safely.	5.1 Machinery is parked in designated safe storage areas. 5.2 Parking brakes and safety locks are applied. 5.3 Tools and accessories are stored in proper storage units. 5.4 Keys and control panels are secured to prevent unauthorized use. 5.5 Storage areas are kept clean and organized.

Range of Variables

Variable	Range (Includes but not limited to):
1. Cleaning tools and chemicals	Cleaning tools 1.1 Brush 1.2 Scraper 1.3 Cloth 1.4 Air blower 1.5 Water hose Cleaning chemicals 1.6 Degreaser 1.7 Detergent 1.8 Parts cleaner 1.9 Rust remover
2. Maintenance tasks	2.1 Engine check 2.2 Hydraulic system check 2.3 Gear oil, hydraulic oil, transmission oil check or replace. 2.4 Filters change 2.5 Greasing 2.6 Cooling system check 2.7 Battery check 2.8 Track or drum inspection 2.9 Brake and control check 2.10 Electrical system check 2.11 Cleaning and leak inspection

Curricular Content Guide

1. Underpinning Knowledge	1.1 Basic concept of cleaning requirements for heavy machinery. 1.2 Purpose and types of cleaning tools, chemicals, and materials used for safe machine cleaning. 1.3 Importance of cleaning operator cabins, controls, windows, drums, buckets, and undercarriage parts for visibility and performance.
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	1.4 Concept of lubrication points, types of lubricants, and why routine lubrication prevents wear and friction. 1.5 Concept of routine maintenance schedules, calibration needs, and the difference between minor and major repairs. 1.6 Importance of proper storage practices. 1.7 Purpose of documentation and maintenance logs for tracking cleaning, lubrication, and repair activities.
2. Underpinning Skills	2.1 Cleaning machinery and components after operation. 2.2 Lubricating moving parts and following maintenance intervals. 2.3 Inspecting components for wear, cracks, leaks, or damage. 2.4 Performing routine maintenance and minor adjustments. 2.5 Identifying unsafe parts and reporting faults promptly. 2.6 Storing machinery, tools, and accessories safely. 2.7 Maintaining records and logs for all maintenance activities.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	The following resources must be provided: 4.1 Workplace (actual or simulated) 4.2 Tools, equipment and facilities appropriate to processes or activity 4.3 Materials relevant to the proposed activity 4.4 Relevant manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 cleaned machinery after operation. 1.2 lubricated moving parts routinely. 1.3 inspected components for wear or damage. 1.4 performed routine maintenance. 1.5 stored machinery and tools safely.
2. Methods of Assessment	Competency should be assessed by:

	2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Heavy Equipment Operation (Road Roller & Excavator)
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S. N.	Major Equipment and Training facilities	Required facilities
1.	Computer/laptop	01
2.	Multimedia projector	01
3.	Road roller	01
4.	Excavator	01
5.	Excavator Attachments (Orange grapple)	01
6.	Blower	02
7.	Air Compressor	02
8.	Air pressure gauge	01
9.	Filter opener set	05
10.	Spanner set	05
11.	Allen key set	05
12.	Socket set	05
13.	Screwdriver set	05
14.	Pliers	05
15.	Hammer	05
16.	Torque wrench	05
17.	Grease gun	05

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Heavy Equipment Operation (Road Roller & Excavator), the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.